

AFLP-based assessment of genetic diversity among nine alfalfa germplasms using bulk DNA templates

A. Segovia-Lerma, R.G. Cantrell, J.M. Conway, and I.M. Ray

Abstract: Improving commercial utilization of perennial *Medicago* collections requires developing approaches that can rapidly and accurately characterize genetic diversity among large numbers of populations. This study evaluated the potential of using amplified fragment length polymorphism (AFLP) DNA markers, in combination with DNA bulking over multiple genotypes, as a strategy for high-throughput characterization of genetic distances (D) among alfalfa (*Medicago sativa* L.) accessions. Bulk DNA templates from 30 genotypes within each of nine well-recognized germplasms (African, Chilean, Flemish, Indian, Ladak, *Medicago sativa* subsp. *falcata*, *Medicago sativa* subsp. *varia*, Peruvian, and Turkistan) were evaluated using 34 primer combinations. A total of 3754 fragments were identified, of which 1541 were polymorphic. The number of polymorphic fragments detected per primer combination ranged from 20 to 85. Pairwise D estimates among the nine germplasms ranged from 0.52 to 1.46 with *M. sativa* subsp. *falcata* being the most genetically dissimilar. Unweighted pair-group method arithmetic average (UPGMA) analysis of the marker data produced two main clusters, (i) *M. sativa* subsp. *sativa* and *M. sativa* subsp. *varia*, and (ii) *M. sativa* subsp. *falcata*. Cluster-analysis results and D estimates among the Chilean, Peruvian, Flemish, and *M. sativa* subsp. *varia* germplasms supported the hypothesis that Peruvian was more similar to original Spanish introductions into Central and South America than Chilean. Hierarchical arrangement of the nine germplasms was supported by their respective geographic, subspecific, and intersubspecific hybrid origins. Subsets of as few as seven highly informative primer pairs were identified that produced comparable D estimates and similar hierarchical arrangements compared with the complete dataset. The results indicate that use of primer-pair subsets for AFLP analysis of bulk DNA templates could serve as a high-throughput system for accurately characterizing genetic diversity among large numbers of alfalfa populations.

Key words: *Medicago sativa*, DNA bulking, genetic distance.

Résumé : L'accroissement de l'emploi commercial des collections de *Medicago* pérennes nécessite le développement d'approches permettant de mesurer rapidement et précisément la diversité génétique au sein d'un grand nombre de populations. Cette étude visait à évaluer l'utilité potentielle des marqueurs AFLP (polymorphisme de longueur des fragments amplifiés) en combinaison avec l'emploi d'échantillons d'ADN provenant de plusieurs génotypes dans le contexte d'une stratégie à haut débit pour la mesure des distances génétiques (D) entre accessions de la luzerne (*Medicago sativa* L.). Des mélanges d'ADN provenant de 30 génotypes au sein de chacune de neuf collections de ressources génétiques bien typées (africaine, chilienne, flamande, ladak, *M. sativa* subsp. *falcata*, *M. sativa* subsp. *varia*, péruvienne et turkistanaise) ont été analysés avec 34 combinaisons d'amorces. Au total, 3754 fragments ont été identifiés dont 1541 étaient polymorphes. Le nombre de fragments polymorphes détectés par paire d'amorces variait entre 20 et 85. Des estimations de D , en prenant les neuf accessions prises deux à deux, variaient entre 0,52 et 1,46 ; le *M. sativa* subsp. *falcata* étant le plus différent des autres. Une analyse UPGMA (« unweighted pair-group method arithmetic average ») des données a produit deux groupes (i) *M. sativa* subsp. *sativa* et *M. sativa* subsp. *varia* et (ii) *M. sativa* subsp. *falcata*. Les résultats de la mesure de D et de l'analyse de groupement parmi les accessions chiliennes, péruviennes, flamandes et le *M. sativa* subsp. *varia* appuient l'hypothèse voulant que les ressources génétiques péruviennes sont plus semblables aux introductions espagnoles originales en Amérique du Sud et centrale que leurs homologues du Chili. Un arrangement hiérarchique des neuf accessions était supporté par leurs origines respectives sur le plan géographique, subsppécifique et de l'hybridation inter-subspécifique. Un sous-ensemble d'aussi peu que sept paires d'amorces très informatives a été identifié, lequel produit des estimés de D et des hiérarchies qui sont comparables à ceux obtenus avec l'ensemble des données. Ces résultats indiquent

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que l'emploi de combinaisons choisies d'amorces AFLP sur des mélanges d'ADN pourrait permettre la caractérisation à haut débit de la diversité génétique au sein d'un grand nombre de populations de la luzerne.

Mots clés : *Medicago sativa*, mélanges d'ADN, distance génétique.

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Introduction

A wealth of genetic variation for many crops is available through gene banks around the world. Efficient use of these resources requires accurate characterization of genetic variation within and among populations. Molecular-marker-based genetic diversity assessments of alfalfa (*Medicago sativa* L.) accessions offers a promising approach to design more effective strategies to use perennial *Medicago* collections. Kidwell et al. (1994a) reported significant positive associations between the genetic diversity of parental alfalfa genotypes and the forage yield of their respective single-cross progenies. In another study, two thirds of alfalfa populations generated from genetically diverse parents had higher forage yields than populations generated from genetically similar parents (Kidwell et al. 1999). Brummer (1999) proposed that identification and maintenance of distinctly divergent populations may be necessary to develop heterotic groups in alfalfa. Molecular markers have proven useful for assigning populations to heterotic groups in other crops (Cheres et al. 2000; Hongtrakul et al. 1997; Melchinger and Gumber 1998; Messmer et al. 1992; Mumm and Dudley 1994).

Different molecular marker types have been used to assess genetic diversity in alfalfa, including isozymes (Quiros and Bauchan 1988), RFLPs (Brummer et al. 1991; Kidwell et al. 1994b), random amplified polymorphic DNA (RAPD) (Ghérardi et al. 1998; Yu and Pauls 1993), and simple sequence repeats (Diwan et al. 1997). Amplified fragment length polymorphisms (AFLPs) (Vos et al. 1995) were also demonstrated to be reproducible, heritable, and highly informative when compared with other marker systems in alfalfa (Barcaccia et al. 1999), similar to results in other crops (Powell et al. 1996; Yee et al. 1999). RFLPs, RAPDs, and AFLPs have been used to identify markers associated with meiotic irregularities in diploid alfalfa (Barcaccia et al. 2000) and winter hardiness traits in tetraploid alfalfa (Brouwer et al. 2000).

Genetic diversity studies require that sufficient markers be used to adequately sample the genome. This is especially true in allogamous polyploids, such as cultivated alfalfa ($2n = 4x = 32$), where populations are highly heterozygous and heterogeneous. The most extensive study to date evaluated genetic diversity of nine alfalfa populations using 180 RFLPs (Kidwell et al. 1994b). This probably represents the minimum number of markers needed, given that approximately 200 markers are required to have a 95% chance of locating at least one marker on each of tetraploid alfalfa's 64 chromosome arms (Sedcole 1977). Brummer et al. (1995) evaluated diversity among six annual *Medicago* species and reported that a minimum of 10 RAPD primers (i.e., 178 markers) were necessary to obtain species groupings that were consistent with traditional taxonomy. The use of fewer primers did not provide expected phylogenetic relationships, indicating that more primers and (or) markers provided a

more accurate picture of species relationships. Greater numbers of markers may be required to obtain accurate phylogenetic relationships among populations within the *M. sativa* subspecies complex.

Most DNA marker-based diversity studies in alfalfa have analyzed individual genotypes from relatively few populations. The ability to discriminate between populations has been hindered by high levels of genetic variation within and between populations, and identification of very few population-specific alleles (Kidwell et al. 1994b; Ghérardi et al. 1998). Estimation of RAPD marker frequencies using randomly sampled genotypes within a population, followed by comparison of marker frequencies between populations, appears to have potential for discriminating between alfalfa cultivars and testing for uniformity among seed lots (Ghérardi et al. 1998). While marker analysis of individual genotypes provides the greatest information about a population's genetic structure, this approach is not practical if large numbers of populations are to be characterized. Hence, the bulking of DNA templates over multiple alfalfa genotypes was proposed (Yu and Pauls 1993). These authors evaluated four populations using RAPD markers and preliminarily demonstrated the usefulness of the DNA-bulking approach. Concerns about DNA bulking were subsequently raised because low-frequency fragments would be obscured, which could reduce the ability to distinguish between populations (Kidwell et al. 1994b). Although this argument has merit, a rigorous examination has not been conducted to determine if biologically relevant genetic distance estimates can be obtained in alfalfa using bulked DNA templates.

If large numbers of populations are to be evaluated, the selected marker system must be accurate, highly informative, cost effective, and suitable for automation. AFLPs fulfill each of these requirements and may enhance the ability to resolve genetic relationships between alfalfa populations. Combining AFLP analysis with a DNA bulking approach may provide an opportunity to develop a high-throughput system for accurately characterizing genetic diversity among large numbers of alfalfa populations. To determine the potential use of this strategy, we evaluated accessions representing nine well-known *M. sativa* germplasm sources. These germplasms provided at least one representative of the three recognized subspecies, *M. sativa* subsp. *falcata*, *M. sativa* subsp. *sativa*, and *M. sativa* subsp. *varia* (Basigalup et al. 1995). Our objectives were as follows: (i) to use a highly informative DNA marker system (e.g., AFLPs) to estimate population-based genetic distances from DNA bulks; (ii) to determine if such diversity estimates were biologically relevant based on agreement between the hierarchical arrangement of the germplasms and their respective geographic, subspecific, and intersubspecific hybrid origins; and (iii) to identify useful subsets of highly informative primers that could accurately characterize genetic distances

Table 1. Accessions used to represent nine alfalfa germplasm sources.

Germplasm designation	Representative accessions (seed lot production date)	Origin of seed source
African	Moapa (pre-1956) Higazzi (pre-1956)	L. Teuber and C. Stanford, University of California at Davis L. Teuber and C. Stanford, University of California at Davis
Chilean	Arizona Chilean (1957) Chilean 21-25 (1950)	B. Melton, New Mexico State University B. Melton, New Mexico State University
Flemish	DuPuits (1960), PI 206103 Alfa (1960), PI 233715	L. Teuber and C. Stanford, University of California at Davis L. Teuber and C. Stanford, University of California at Davis
Indian	FC23631 (pre-1960) FC32174 (pre-1960)	L. Teuber and C. Stanford, University of California at Davis L. Teuber and C. Stanford, University of California at Davis
Ladak	Ladak 1 (pre-1950) Ladak 2 (pre-1950)	E.T. Bingham, University of Wisconsin at Madison E.T. Bingham, University of Wisconsin at Madison
<i>M. sativa</i> subsp. <i>falcata</i>	WISFAL (1993), PI 560333	E.T. Bingham, University of Wisconsin at Madison
<i>M. sativa</i> subsp. <i>varia</i>	Grimm (1960) Cossack (1951)	L. Teuber and C. Stanford, University of California at Davis E.T. Bingham, University of Wisconsin at Madison
Peruvian	Hairy Peruvian (1959) Coastal Hairy Peruvian (1959)	B. Melton, New Mexico State University B. Melton, New Mexico State University
Turkistan	Turkistan-Samarkand (pre-1950) Turkistan-Tokmak (pre-1950)	E.T. Bingham, University of Wisconsin at Madison E.T. Bingham, University of Wisconsin at Madison

among these populations for future development of a high-throughput system.

Materials and methods

Plant material and DNA preparation

Thirty genotypes from each of nine alfalfa germplasm sources — African, Chilean, Flemish, Indian, Ladak, *M. sativa* subsp. *falcata*, *M. sativa* subsp. *varia*, Peruvian, and Turkistan (Barnes et al. 1977) — were analyzed in this study. The *M. sativa* subsp. *falcata* germplasm was represented by a single broad-based accession, WISFAL (Bingham 1993). The other eight germplasms were represented by 15 plants from each of 2 relic accessions derived primarily from the 1940–1960 era (Table 1). Relic accessions were used to increase the likelihood of germplasm purity, because historical records indicate that as time has progressed the use of recombination nurseries to hybridize two or more alfalfa germplasms has become increasingly common (Barnes et al. 1988). Outcrossing and contamination of more recently derived populations, therefore, may have had a greater chance of occurring, as alluded to by Kidwell et al. (1994b).

Total DNA was extracted from 10–15 immature trifoliate leaves from each of the 30 plants representing each germplasm ($n = 270$), according to Kidwell and Osborn (1992). Purified DNA from each genotype was quantified. Nine genomic bulked-DNA solutions were prepared by mixing equal quantities (i.e., 500 ng/plant) of DNA from each of the 30 plants representing each germplasm. All samples were stored at -20°C .

AFLP analysis

The AFLP protocol described by Vos et al. (1995) was followed with some modifications. All reagents were supplied by Perkin Elmer Applied Biosystems (Foster City, Calif.) except for restriction enzymes and their buffers, which were supplied by New England Biolabs (Beverly,

Mass.). Restriction–ligation reactions for each germplasm were prepared in a reaction volume of 11 μL that contained 500 ng of bulked genomic DNA in 5.5 μL of water, 1 μL of $10\times$ T4 DNA ligase buffer with ATP, 1 μL of 0.5 M NaCl, 0.5 μL of 100 μg BSA/mL, 1 μL of a $10\times$ master mix, 1 μL of an *MseI* adaptor, and 1 μL of an *EcoRI* adaptor (Table 2). The $10\times$ master mix contained 2.25 μL of sterile distilled water, 1 μL of $10\times$ T4 DNA ligase buffer with ATP, 1 μL of 0.5 M NaCl, 0.5 μL of 1 mg BSA/mL, 2.5 μL of 4 U *MseI*/ μL , 2.5 μL of 20 U *EcoRI*/ μL , and 0.25 μL of 40 U T4 DNA ligase/ μL . The master mix tube was vortexed, centrifuged at 2500 rpm ($600\times g$) for 10 s, and kept on ice until ready to use. Before adding adaptor pairs, the tubes were briefly centrifuged at 4000 rpm ($1400\times g$). Reactions were incubated overnight at 37°C in a Perkin Elmer thermal cycler (model 9600, Perkin Elmer Applied Biosystems). Restriction–ligation reactions were diluted to a final volume of 200 μL with $0.1\times$ TE (1 mM Tris–HCl and 0.1 mM EDTA) buffer. Preselective and selective amplifications were performed according to the manufacturer's recommendations using the primers described in Table 2. Eight fluorescently labeled *EcoRI* selective primers (designated A–H for nucleotides selective for AC, AG, CA, CC, CG, CT, GC, and GG, respectively) were each used in combination with eight *MseI* selective primers (designated 1–8 for nucleotides selective for AA, AC, AG, AT, TA, TC, TG, and TT, respectively) to amplify bulked DNA templates for each of the nine germplasms. All amplification reactions were carried out in 200- μL tubes in a Perkin Elmer thermocycler (model 9600). The PCR products were mixed with an equal volume of loading buffer (60% deionized formamide, 5.8 mM EDTA, 11.7 mg blue dextran/mL, and 17% GeneScan-500 Rox size standards (PN401734, Perkin Elmer Applied Biosystems)) and were denatured for 3 min at 94°C and then placed on ice. Two microlitres of each reaction mix were electrophoresed on a 5% w/v long Ranger (FMC Bioproducts, Rockland, Me.) denaturing polyacrylamide gel with 6 M urea that was 36 cm in length. Gels were run for

Table 2. Oligonucleotide sequences for adaptors and primers used for AFLP analyses.

Restriction enzyme	Sequence
Adaptors	
<i>Eco</i> RI	5'-CTCGTAGACTGCGTACC-3' 3'-CATCTGACGCATGGTTAA-5'
<i>Mse</i> I	5'-GACGATGAGTCCTGAG-3' 3'-TACTCAGGACTCAT-5'
Preselective primers	
<i>Eco</i> RI	5'-GACTGCGTACCAATTCA-3'
<i>Mse</i> I	5'-GATGAGTCCTGAGTAAC-3'
Selective primers	
<i>Eco</i> RI	5'-GACTGCGTACCAATTCA-NN-3'*
<i>Mse</i> I	5'-GATGAGTCCTGAGTAAC-XX-3'†

*NN represents the selective dinucleotides AC, AG, CA, CC, CG, CT, GC, and GG.

†XX represents the selective dinucleotides AA, AC, AG, AT, TA, TC, TG, and TT.

Table 3. Polymorphism number and rate for 34 AFLP primer pairs used to amplify bulked genomic DNA templates from 9 alfalfa germplasms.

Primer pair*	Selective nucleotides (<i>Eco</i> RI/ <i>Mse</i> I)	Total no. of fragments	No. of polymorphic fragments	Polymorphism rate (%)
A1	AC/AA	81	21	26
A2	AC/AC	109	45	41
A3	AC/AG	74	22	30
B1	AG/AA	96	20	21
B2	AG/AC	114	53	47
B4	AG/AT	115	22	19
B5	AG/TA	125	46	37
B6	AG/TC	109	47	43
B8	AG/TT	117	33	28
C1	CA/AA	158	72	47
C2	CA/AC	119	55	46
C3	CA/AG	159	80	50
C4	CA/AT	171	76	44
C5	CA/TA	127	51	40
C6	CA/TC	147	66	45
C7	CA/TG	151	85	56
C8	CA/TT	147	57	39
D1	CC/AA	114	54	47
D3	CC/AG	82	44	54
D4	CC/AT	80	33	41
D5	CC/TA	56	26	46
D6	CC/TC	53	23	43
D7	CC/TG	85	48	57
D8	CC/TT	102	43	42
E1	CG/AA	101	37	37
E2	CG/AC	72	20	28
E4	CG/AT	85	32	38
F2	CT/AC	102	44	43
F3	CT/AG	129	51	40
F4	CT/AT	105	28	27
F5	CT/TA	112	36	32
F6	CT/TC	109	58	53
F8	CT/TT	126	56	44
G1	GC/AA	122	57	47
Total		3754	1541	40
Mean		110	45	40

*Letters A–H designate *Eco*RI primers and numbers 1–8 designate *Mse*I primers as described in the Materials and methods.

Table 4. Number and distribution of AFLP markers within and among nine alfalfa germplasms.

		Number of markers							
Germplasm	Accession specific	Common between two or more germplasms*							Total [†]
		2	3	4	5	6	7	8	
African	16	40	51	62	68	82	129	305	753
Chilean	10	33	45	47	65	82	134	322	739
Flemish	7	39	51	57	74	96	149	343	817
Indian	11	22	28	48	60	69	127	323	689
Ladak	17	65	60	69	69	88	145	346	859
<i>M. sativa</i> subsp. <i>falcata</i>	213	88	55	70	56	68	101	161	812
<i>M. sativa</i> subsp. <i>varia</i>	13	30	40	62	67	92	137	341	783
Peruvian	11	20	41	50	52	88	139	335	737
Turkistan	25	45	37	71	59	85	129	332	783
Mean	36	42	45	60	63	83	132	312	775

*2213 fragments were present in all nine germplasms (i.e., monomorphic).

[†]Total number of markers identified within each germplasm.

Table 5. AFLP-based genetic distance estimates among nine alfalfa germplasms using bulked DNA templates.

	Chilean	Flemish	Indian	Ladak	<i>M. sativa</i> subsp. <i>falcata</i>	<i>M. sativa</i> subsp. <i>varia</i>	Peruvian	Turkistan	Mean
African	0.69	0.60	0.73	0.66	1.40	0.67	0.66	0.70	0.76
Chilean	—	0.58	0.63	0.63	1.43	0.60	0.57	0.67	0.73
Flemish	—	—	0.60	0.54	1.16	0.52	0.54	0.60	0.64
Indian	—	—	—	0.67	1.46	0.59	0.60	0.68	0.74
Ladak	—	—	—	—	1.12	0.52	0.63	0.60	0.67
<i>M. sativa</i> subsp. <i>falcata</i>	—	—	—	—	—	1.24	1.37	1.30	1.31
<i>M. sativa</i> subsp. <i>varia</i>	—	—	—	—	—	—	0.56	0.66	0.67
Peruvian	—	—	—	—	—	—	—	0.60	0.69
Turkistan	—	—	—	—	—	—	—	—	0.72

7 h in 1× TBE (89 mM Tris, 89 mM borate, 2 mM EDTA) at 1680 V constant voltage using an ABI 377 DNA sequencer (Perkin Elmer Applied Biosystems).

Reproducibility of the AFLP fingerprints was assessed by generating two independent restriction–ligation and preselective amplification reactions for all nine germplasm genomic DNA bulks. The two preselectively amplified reactions for each germplasm were amplified using two selective primer combinations, B1 and B4 (*Eco*RI-AG with *Mse*I-AA, and *Eco*RI-AG with *Mse*I-AT, respectively). The two independent selective amplification products representing each germplasm were loaded pairwise on a gel and were electrophoresed. Repeatability of the AFLP profile for each germplasm was estimated based on the number of fragments in common between paired reactions as compared with the total number of DNA fragments in the two reactions.

Data collection and analysis

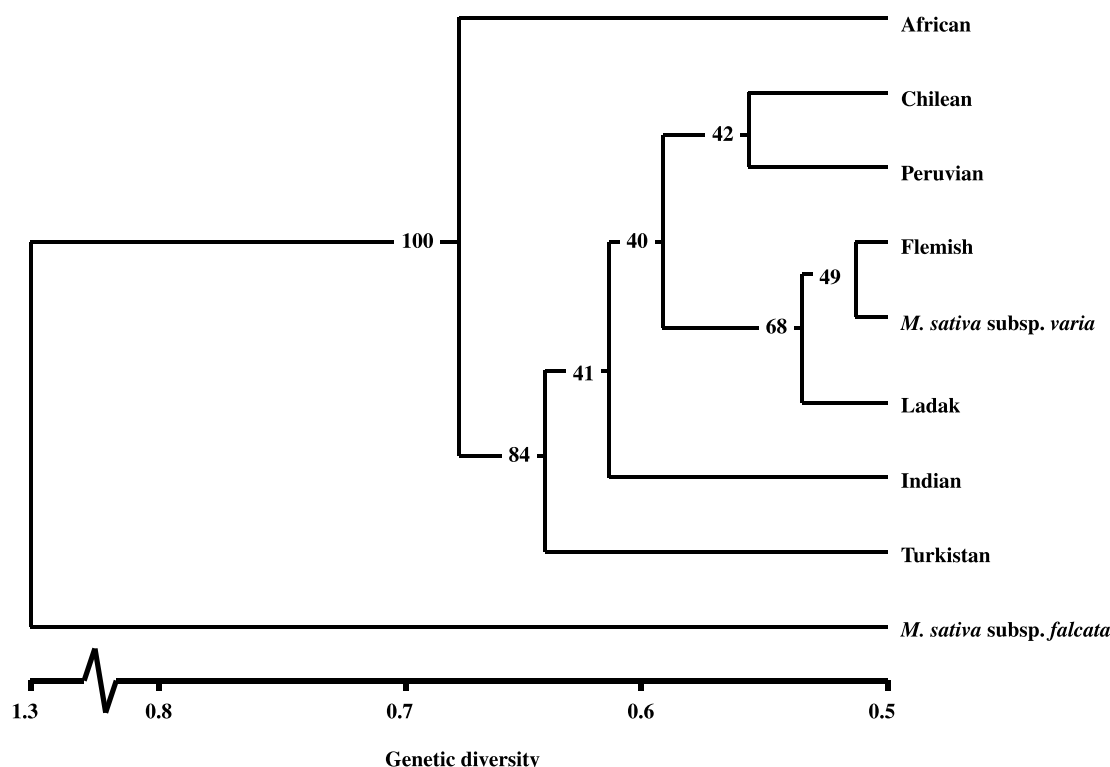
Of the 64 primer-pair combinations evaluated, 30 failed to amplify one or more of the bulked DNA templates and were not included in the analysis. For each of the remaining 34 primer pairs, the number of polymorphic fragments was determined by visually evaluating the electropherogram produced by the GeneScan software program (version 2.1.0) in which all nine templates were overlaid. Only clearly defined peaks that were absent in at least one germplasm were scored as 1 for present and 0 for absent and entered into a

data matrix. Monomorphic fragments were excluded from data analysis. Fragment size was determined by the GeneScan software using the local Southern method. Fragments were considered to be different when separated by at least ±1 base pair. The 1-0 data matrix was analyzed using NTSYSpc version 2.02j (Rohlf 1998) and 36 pairwise genetic similarity estimates were produced. Similarity estimates were calculated based on Jaccard's coefficient of similarity (J_{ij}) where $J_{ij} = a/(n - d)$, and a is the number of fragments in common between two germplasms, i and j , n is the total number of fragments scored, and d is the number of fragments absent in both germplasms. Similarity estimates were converted to genetic distance (D) where $D = -\ln J$, according to Swofford and Olson (1990). Diversity patterns were represented in the form of a dendrogram that was generated by subjecting the genetic distance matrix to unweighted pair-group method arithmetic average (UPGMA) cluster analysis with NTSYSpc software (Rohlf 1998). Support for clusters was determined by bootstrap analysis (i.e., repeated sampling with replacement) of 1000 permuted datasets that were generated using WinBoot software (Yap and Nelson 1996).

Results and discussion

The 34 primer pairs used in this study amplified 1541 polymorphic and 2213 monomorphic fragments across the nine alfalfa germplasms. The repeatability of AFLP frag-

Fig. 1. UPGMA dendrogram based on genetic distances as estimated by 1541 AFLPs among nine alfalfa populations. Numbers at branch points indicate support for groupings to the right of the number; values are percent of bootstrap datasets that exhibited the cluster.



ments exceeded 98% for each of the nine germplasms based on the two independent restriction–ligation reactions of each population that were amplified using two selective primer pairs (data not shown). The results indicate that DNA bulks were reproducibly amplified and that most polymorphisms represented true genetic differences among the populations. The number of polymorphic fragments generated per primer pair ranged from 20 to 85 with an average of 45 (Table 3). Polymorphism rates ranged from 19 to 57% with a mean of 40%. The C1–C8 primer pairs were highly informative and accounted for 542 markers. Six other primer pairs, B2, D1, F3, F6, F8, and G1, also detected more than 50 AFLPs each for an additional 329 markers. These 14 primer pairs, or subsets of them, should be useful in other marker-based genetic diversity studies of the *M. sativa* complex.

The total number of polymorphic fragments identified within each of the nine germplasms ranged from 689 in Indian to 859 in Ladak (Table 4). Approximately 75% of the markers were present in at least five germplasms. Three hundred and thirty-one AFLP alleles were classified as germplasm specific because they occurred in only a single accession. With the exception of the 213 fragments identified in *M. sativa* subsp. *falcata*, the frequency of “accession specific” fragments was within the 2% margin of error detected in the repeatability assay. More extensive sampling within each of the bulked templates would be required to estimate the proportion of fragments that are truly accession specific, as opposed to those that represent PCR artifacts or reflect sampling variation from heterogeneous templates.

Genetic distance values based on pairwise comparisons among the nine alfalfa germplasms ranged from 0.52 to 1.46 (Table 5). *Medicago sativa* subsp. *falcata* was the most genet-

ically dissimilar among the germplasms studied (D mean = 1.31). Pairwise D estimates among the other eight germplasms ranged from a low of 0.52 for Flemish vs. *M. sativa* subsp. *varia* to a high of 0.73 for African vs. Indian. The dendrogram based on UPGMA cluster analysis (Fig. 1) grouped the nine germplasms into two main clusters of (i) *M. sativa* subsp. *sativa* and *M. sativa* subsp. *varia*, and (ii) *M. sativa* subsp. *falcata*. Within the *M. sativa* subsp. *sativa* and *M. sativa* subsp. *varia* cluster, hierarchical arrangement of the Chilean, Flemish, Indian, Ladak, *M. sativa* subsp. *varia*, and Peruvian germplasms were supported by their respective geographic, subspecific, and intersubspecific hybrid origins. The Chilean and Peruvian populations, which were of European and northwestern African origin but independently subjected to several hundred years of selection in Central and South America, grouped together. These two populations were hierarchically adjacent to another cluster of Flemish, *M. sativa* subsp. *varia*, and Ladak. Ladak was hierarchically adjacent to *M. sativa* subsp. *varia* and Flemish, and Indian, but was most similar to *M. sativa* subsp. *varia*, likely reflecting their intersubspecific hybrid nature. Both Ladak and *M. sativa* subsp. *varia* segregated for purple and yellow flowers indicative of *M. sativa* subsp. *sativa* and *M. sativa* subsp. *falcata*, respectively, as well as variegated (hybrid) flowers. The frequency of yellow and variegated flowers, however, was relatively low in both populations, suggesting that they contained predominately *M. sativa* subsp. *sativa* germplasm. The placement of Ladak adjacent to Indian within the dendrogram agreed with their proximate geographic origins of northern (Himalayan) and southern (subtropical) regions of India. The broad-based *M. sativa* subsp. *falcata* germplasm used in this study was

derived from accessions collected throughout Turkey, Switzerland, Yugoslavia, and the former U.S.S.R. (Bingham 1975, 1993).

These results allowed us to test a hypothesis reported by Barnes et al. (1977), who proposed that Peruvian might be more similar to original Spanish introductions into Central and South America than Chilean. Pairwise D estimates (Table 5) between Chilean and Flemish (0.58), Chilean and *M. sativa* subsp. *varia* (0.60), Peruvian and Flemish (0.54), and Peruvian and *M. sativa* subsp. *varia* (0.56) indicated that Peruvian was more similar to the European accessions than Chilean. These results were also supported by bootstrap analysis. Clusters containing Peruvian with Flemish and (or) *M. sativa* subsp. *varia* appeared in 33% of the individual bootstrap trees/replicates, but Chilean clustered with Flemish and (or) *M. sativa* subsp. *varia* in only 0.6% of the replicates.

We evaluated the feasibility of using fewer primers to obtain reliable D estimates and hierarchical arrangements, as a means to more efficiently obtain marker-based diversity estimates on larger numbers of populations in future research. Assuming the D estimates and population hierarchical arrangement based on the complete dataset to be reasonably accurate, the marker data were partitioned to identify primer-pair subsets that could maintain these relationships. Given that the number of factorial combinations associated with 34 primer pairs is prohibitively large (2.3×10^9), data subsets were created based on primer-pair informativeness and sampling of diverse restriction sites. Results of the analyzed subsets were subsequently compared with the complete dataset. Cluster analyses and rank correlations (r_s) based on D estimates indicated that the 14 most informative primers (C1–C8, B2, D1, F3, F6, F8, and G1) were highly correlated with the complete dataset ($r_s = 0.88$, $n = 36$) and essentially preserved the dendrogram arrangement with only an inversion between Indian and Turkistan (data not shown). The C1–C8 subset also showed high rank correlations ($r_s = 0.88$, $n = 36$); however, the cluster of Flemish, *M. sativa* subsp. *varia*, and Ladak was more closely associated with African than with the cluster of Chilean and Peruvian. Two other subsets containing seven primer pairs (A2, C1, D4, E1, E4, F8, and G1, and A2, B5, C3, D7, E4, F6, and G1) produced high rank correlations ($r_s = 0.94$, $n = 36$ for both subsets) and identical hierarchical arrangements with the complete dataset, although dendrogram branch lengths differed slightly with each subset. These results suggest that fewer highly informative primer pairs can be used to provide accurate diversity estimates among populations.

Genetic distance estimates among the nine germplasms evaluated in our study were lower than those reported by Kidwell et al. (1994b). This could reflect that the seed stocks used to represent the germplasm sources differed in the two studies. The use of DNA bulks in our study may also have reduced marker-based genetic distances compared with evaluation of individual genotypes within germplasms, because it favored the detection of “consensus” fragments (i.e., common alleles) in each population (Yu and Pauls 1993). Consequently, lower-frequency or unique alleles present in only a few genotypes may have been diluted to such an extent that they were not amplified efficiently. Some allelic information, therefore, was likely lost as a result of

DNA bulking. From a population perspective, the DNA-bulking method permits sampling of a greater number of individuals in heterogeneous heterozygous populations. This approach may more accurately reflect a population's general genetic composition compared with evaluation of fewer individual genotypes. It also permits greater population numbers to be evaluated with comparable resources.

The feasibility of using bulk DNA templates in future marker-based assessments of genetic diversity in alfalfa was supported by the observation that hierarchical patterns of diversity among the nine germplasms were associated with their geographic, subspecific, and intersubspecific hybrid origins. Subsets of primer pairs were also identified that essentially preserved population hierarchical arrangement and produced D estimates with high rank correlations when compared with the complete dataset. Use of these primer subsets for automated AFLP analysis of bulk DNA templates could be used as a high-throughput system for accurately characterizing genetic diversity among large numbers of alfalfa populations. This information should prove useful in designing strategies to more efficiently manipulate heterosis in alfalfa. This approach could subsequently be refined to include individual genotype analysis for more detailed characterization of specific populations of interest.

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References

- Barnes, D.K., Bingham, E.T., Murphy, R.P., Hunt, O.J., Beard, D.F., Skrdla, W.H., and Teuber, L.R. 1977. Alfalfa germplasm in the United States: genetic vulnerability, use, improvement, and maintenance. United States Department of Agriculture – Agricultural Research Service Technical Bulletin No. 1571. Washington, D.C.
- Barnes, D.K., Golpen, B.P., and Baylor, J.E. 1988. Highlights in the U.S.A. and Canada. In *Alfalfa and alfalfa improvement*. Edited by A.A. Hanson, D.K. Barnes, and R.R. Hill, Jr. Agronomy Monographs No. 29. American Society of Agronomy, Crop Science Society of America, and Soil Science Society of America, Madison, Wis. pp. 1–24.
- Barcaccia, G., Albertini, E., Tavoletti, S., Falcinelli, M., and Veronesi, F. 1999. AFLP fingerprinting in *Medicago* spp. Its development and application in linkage mapping. *Plant Breed.* **118**: 335–340.
- Barcaccia, G., Albertini, E., Rosellini, D., Tavoletti, S., and Veronesi, F. 2000. Inheritance and mapping of $2n$ -egg production in diploid alfalfa. *Genome*, **43**: 528–537.
- Basigalup, D.H., Barnes, D.K., and Stucker, R.E. 1995. Development of a core collection for perennial *Medicago* plant introductions. *Crop Sci.* **35**: 1163–1168.
- Bingham, E.T. 1975. Registration of alfalfa germplasm from cultivated $\times$ wild hybrids. *Crop Sci.* **15**: 889.
- Bingham, E.T. 1993. Registration of WISFAL alfalfa (*Medicago sativa* ssp. *falcata*) tetraploid germplasm derived from diploids. *Crop Sci.* **33**: 217–218.

- Brouwer, D.J., Duke, S.H., and Osborn, T.C. 2000. Mapping genetic factors associated with winter hardiness, fall growth, and freezing injury in autotetraploid alfalfa. *Crop Sci.* **40**: 1387–1396.
- Brummer, E.C. 1999. Capturing heterosis in forage crop cultivar development. *Crop Sci.* **39**: 943–954.
- Brummer, E.C., Kochert, G., and Bouton, J.H. 1991. RFLP variation in diploid and tetraploid alfalfa. *Theor. Appl. Genet.* **83**: 89–96.
- Brummer, E.C., Bouton, J.H., and Kochert, G. 1995. Analysis of annual *Medicago* species using RAPD markers. *Genome*, **38**: 362–367.
- Cheres, M.T., Miller, J.F., Crane, J.M., and Knapp, S.J. 2000. Genetic distance as a predictor of heterosis and hybrid performance within and between heterotic groups in sunflower. *Theor. Appl. Genet.* **100**: 889–894.
- Diwan, N., Bhagwat, A.A., Baughan, G.R., and Cregan, P.B. 1997. Simple sequence repeat DNA markers in alfalfa and perennial and annual *Medicago* species. *Genome*, **40**: 887–895.
- Ghérardi, M., Mangin, B., Goffinet, B., Bonnet, D., and Huguet, T. 1998. A method to measure genetic distance between allogamous populations of alfalfa (*Medicago sativa*) using RAPD markers. *Theor. Appl. Genet.* **96**: 406–412.
- Hongtrakul, V., Huestis, G.M., and Knapp S.J. 1997. Amplified fragment length polymorphisms as a tool for DNA fingerprinting sunflower germplasm: genetic diversity among oilseed inbred lines. *Theor. Appl. Genet.* **95**: 400–407.
- Kidwell, K.K., and Osborn, T.C. 1992. Simple plant DNA isolation procedures. *In* Plant genomes: methods for genetic and physical mapping. *Edited by* J. Beckman and T.C. Osborn. Kluwer Academic Publishers, Dordrecht, Netherlands. pp 1–13.
- Kidwell, K.K., Woodfield, D.R., Bingham, E.T., and Osborn, T.C. 1994a. Molecular marker diversity and yield of isogenic 2x and 4x single crosses of alfalfa. *Crop Sci.* **34**: 784–788.
- Kidwell, K.K., Austin, D.F., and Osborn T.C. 1994b. RFLP evaluation of nine *Medicago* accessions representing the original germplasm sources for North American alfalfa cultivars. *Crop Sci.* **34**: 230–236.
- Kidwell, K.K., Hartweck, L.M., Yandell, B.S., Crump, P.M., Brummer, J.E., Moutray, J., and Osborn, T.C. 1999. Forage yields of alfalfa populations derived from parents selected on the basis of molecular marker diversity. *Crop Sci.* **39**: 223–227.
- Melchinger, A.E., and Gumber, R.K. 1998. Overview of heterosis and heterotic groups in agronomic crops. *In* Concepts and breeding of heterosis in crop plants. *Edited by* K.R. Lamkey and J.E. Staub. Special Publication No. 25. Crop Science Society of America, Madison, Wis. pp. 29–44.
- Messmer, M.M., Melchinger, A.E., Boppenmaier, J., Brunklaus-Jung, E., and Herrmann, R.G. 1992. Relationships among early European maize inbreds. I. Genetic diversity among flint and dent lines revealed by RFLPs. *Crop Sci.* **32**: 1301–1309.
- Mumm, R.H., and Dudley, J.W. 1994. A classification of 148 U.S. maize inbreds. I. Cluster analysis based on RFLPs. *Crop Sci.* **34**: 842–851.
- Powell, W., Morgante, M., Andre, C., Hanafey, M., Vogel, J., Tingey, S., and Rafalski, A. 1996. The comparison of RFLP, RAPD, AFLP, and SSR (microsatellite) markers for germplasm analysis. *Mol. Breed.* **2**: 225–238.
- Quiros, C.F., and Baughan, G.R. 1988. The genus *Medicago* and the origin of the *Medicago sativa* complex. *In* Alfalfa and alfalfa improvement. *Edited by* A.A. Hanson, D.K. Barnes, and R.R. Hill, Jr. Agronomy Monographs No. 29. American Society of Agronomy, Crop Science Society of America, and Soil Science Society of America, Madison, Wis. pp. 93–124.
- Rohlf, F.J. 1998. NTSYSpc version 2.02j. Numerical taxonomy and multivariate analysis system. Exeter Software, Setauket, N.Y.
- Sedcole, J.R. 1977. Number of plants necessary to recover a trait. *Crop Sci.* **17**: 667–668.
- Swofford, D.L., and Olson, G.J. 1990. Phylogeny reconstruction. *In* Molecular systematics. *Edited by* D.M. Hillis and C. Mortiz. Sinauer Associates, Sunderland, Mass. pp. 411–501.
- Vos, P., Hogers, R., Bleeker, M., Reijans, M., van de Lee, T., Hornes, M., Frijters, A., Pot, J., Peleman, J., Kuiper, M., and Zabeau, M. 1995. AFLP: a new technique for DNA fingerprinting. *Nucleic Acids Res.* **23**: 4407–4414.
- Yap, I.V., and Nelson, R. 1996. WinBoot: A program for bootstrap analysis of binary data to determine the confidence limits of UPGMA-based dendrograms. International Rice Research Institute, Manila, Philippines.
- Yee, E., Kidwell, K.K., Sills, G.R., and Lumpkin, T.A. 1999. Diversity among selected *Vigna angularis* accessions on the basis of RAPD and AFLP markers. *Crop Sci.* **39**: 268–275.
- Yu, K., and Pauls, K.P. 1993. Rapid estimation of genetic relatedness among heterogeneous populations of alfalfa by random amplification of bulked genomic DNA samples. *Crop Sci.* **86**: 788–794.